NET Mathematics Formula and Trick Book

1. Number System and Algebra Basics

Laws of Indices

$$a^m a^n = a^{m+n}$$
$$\frac{a^m}{a^n} = a^{m-n}, \qquad a \neq 0$$
$$(a^m)^n = a^{mn}$$
$$(ab)^n = a^n b^n$$
$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$
$$a^0 = 1, \qquad a^{-n} = \frac{1}{a^n}$$
$$a^{1/n} = \sqrt[n]{a}, \qquad a^{m/n} = \sqrt[n]{a^m}$$

Surds

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$
$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$
$$(a + b\sqrt{c})(a - b\sqrt{c}) = a^2 - b^2 c$$
$$\frac{1}{a+\sqrt{b}} = \frac{a-\sqrt{b}}{a^2 - b}$$

Basic Factorization

$$a^2 - b^2 = (a-b)(a+b)$$
$$a^2 + 2ab + b^2 = (a+b)^2$$
$$a^2 - 2ab + b^2 = (a-b)^2$$
$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$
$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$
$$x^2 + (a+b)x + ab = (x+a)(x+b)$$

Absolute Value

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$
$$|x-a| < b \Rightarrow a-b < x < a+b, \qquad b > 0$$
$$|x-a| > b \Rightarrow x < a-b \text{ or } x > a+b$$
$$|x-a| = b \Rightarrow x = a+b \text{ or } x = a-b$$

Algebra Tricks

- For $|x-a| < b$, write interval directly as $(a-b, a+b)$.
- For $|x-a| = b$, expect two roots unless $b = 0$.
- Rationalize before matching options.
- Use factorization before expansion.
- If expression contains $x + \frac{1}{x}$, square it:

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}.$$

2. Logarithms and Exponents

Log Laws

$$\log_a(xy) = \log_a x + \log_a y$$
$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$
$$\log_a(x^n) = n \log_a x$$
$$\log_a a = 1, \qquad \log_a 1 = 0$$
$$a^{\log_a x} = x$$
$$\log_a x = \frac{\log_b x}{\log_b a}$$

Domains

$$\log_a x \text{ exists if } x > 0,\ a > 0,\ a \neq 1$$

Exponential Equations

$$a^x = a^y \Rightarrow x = y$$
$$a^x = b \Rightarrow x = \log_a b$$

Example:

$$2^{x+1} = 16 = 2^4 \Rightarrow x + 1 = 4 \Rightarrow x = 3$$

Tricks

- Same base on both sides means equate powers.
- $\log a + \log b = \log(ab)$.
- $\log a - \log b = \log(a/b)$.
- Always check log argument > 0.
- For $\log_2 x = 5$, immediately write $x = 2^5 = 32$.

3. Sets, Relations, Functions

Set Operations

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$
$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$
$$A - B = \{x : x \in A,\ x \notin B\}$$
$$A - B = A \cap B'$$
$$(A \cup B)' = A' \cap B'$$
$$(A \cap B)' = A' \cup B'$$

Counting Formulae

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$n(A - B) = n(A) - n(A \cap B)$$
$$n(A \cup B \cup C) = n(A) + n(B) + n(C)$$
$$-n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

Relations

$$R \subseteq A \times B$$

Reflexive:
$$(a, a) \in R \quad \forall a \in A$$

Symmetric:
$$(a, b) \in R \Rightarrow (b, a) \in R$$

Transitive:
$$(a, b), (b, c) \in R \Rightarrow (a, c) \in R$$

Functions

$$f : A \to B$$

Domain = input set. Codomain = target set. Range = actual output set.

$$(g \circ f)(x) = g(f(x))$$
$$(f \circ g)(x) = f(g(x))$$

Usually:
$$g \circ f \neq f \circ g$$

Inverse Function

Method:
$$y = f(x)$$

Solve for x, then replace y by x.
Example:
$$f(x) = 3x + 5$$
$$y = 3x + 5 \Rightarrow x = \frac{y - 5}{3}$$
$$f^{-1}(x) = \frac{x - 5}{3}$$

Domain Restrictions

$$\sqrt{x - a} \Rightarrow x - a \geq 0$$
$$\frac{1}{x - a} \Rightarrow x \neq a$$
$$\log(x - a) \Rightarrow x - a > 0$$
$$\sqrt{\frac{x - a}{x - b}} \Rightarrow \frac{x - a}{x - b} \geq 0, \quad x \neq b$$

Tricks

- In $A - B$, remove from A only.
- For $g \circ f$, apply f first, then g.
- For inverse, do not take reciprocal unless function itself is reciprocal.
- Domain questions usually hide denominator and square-root restrictions.

4. Groups and Binary Operations

Binary Operation

A binary operation $*$ on G:
$$* : G \times G \to G$$

Closure:
$$a, b \in G \Rightarrow a * b \in G$$

Group Conditions

Closure:
$$a * b \in G$$

Associativity:
$$(a * b) * c = a * (b * c)$$

Identity:
$$a * e = e * a = a$$

Inverse:
$$a * a^{-1} = a^{-1} * a = e$$

Abelian Group

$$a * b = b * a$$

Tricks

- Closure checks whether result remains inside the set.
- Identity leaves every element unchanged.
- Inverse depends on operation, not always $1/a$.
- Associativity is grouping; commutativity is order.

5. Complex Numbers

Basic Form

$$z = a + bi$$
$$\Re(z) = a, \qquad \Im(z) = b$$

Powers of i

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$
$$i^n = i^{n \pmod 4}$$

Conjugate and Modulus

$$\overline{a+bi} = a - bi$$
$$|a+bi| = \sqrt{a^2+b^2}$$
$$z\bar{z} = |z|^2$$
$$|zw| = |z||w|$$
$$\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$$

Division

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{(c+di)(c-di)}$$
$$= \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$$

Argument

$$\arg(z) = \tan^{-1}\frac{b}{a}$$

Check quadrant before final angle.

Polar Form

$$z = r(\cos\theta + i\sin\theta)$$
$$r = |z|, \qquad \theta = \arg(z)$$

De Moivre's Theorem

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$
$$z^n = r^n(\cos n\theta + i\sin n\theta)$$

Cube Roots of Unity

$$1, \omega, \omega^2$$
$$\omega^3 = 1$$
$$1 + \omega + \omega^2 = 0$$
$$\omega^4 = \omega, \qquad \omega^5 = \omega^2$$

Tricks

- For i^n, divide n by 4; use remainder.
- For ω^n, reduce n modulo 3.
- For complex division, multiply by conjugate.
- If only real part is asked, stop after real numerator is found.
- Do not split $\frac{a+bi}{c+di}$ into $\frac{a}{c} + \frac{b}{d}i$.

6. Matrices and Determinants

Order

If matrix has m rows and n columns:

$$\text{order} = m \times n$$

Matrix Multiplication

If:

$$A : m \times n, \qquad B : n \times p$$

then:

$$AB : m \times p$$

Middle dimensions must match.

2 by 2 Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

3 by 3 Determinant

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Inverse of 2 by 2 Matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

provided:

$$ad - bc \neq 0$$

Singular and Non-Singular

$$A \text{ singular} \Rightarrow \det(A) = 0$$
$$A \text{ non-singular} \Rightarrow \det(A) \neq 0$$

Determinant Properties

$$\det(AB) = \det(A)\det(B)$$
$$\det(A^{-1}) = \frac{1}{\det(A)}$$
$$\det(A^T) = \det(A)$$
$$\det(kA) = k^n \det(A)$$

where A is $n \times n$.
For 2×2:

$$\det(kA) = k^2 \det(A)$$

Trace

$$\text{tr}(A) = \text{sum of principal diagonal elements}$$

Rank

Rank = maximum number of linearly independent rows or columns.

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \Rightarrow \text{rank} = 1$$

Tricks

- Determinant is $ad - bc$, not $bc - ad$.
- If two rows are proportional, determinant is zero.
- If a row or column is all zero, determinant is zero.
- Singular means determinant zero, not "has zero entries."
- For AB, order is outer dimensions.

7. Quadratic Equations

Standard Form

$$ax^2 + bx + c = 0, \qquad a \neq 0$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant

$$D = b^2 - 4ac$$
$$D > 0: \text{ real unequal roots}$$
$$D = 0: \text{ real equal roots}$$
$$D < 0: \text{ complex roots}$$

Sum and Product

If roots are α, β:

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Equation from Roots

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Reciprocal Roots

If:

$$ax^2 + bx + c = 0$$

has roots α, β, then equation with roots $1/\alpha, 1/\beta$ is:

$$cx^2 + bx + a = 0$$

Transformed Roots

Roots $\alpha + k, \beta + k$: put $x = y - k$.
Roots $k\alpha, k\beta$: put $x = \frac{y}{k}$.

Tricks

- Equal roots means $D = 0$.
- Sum of roots has minus sign.
- Product of roots has no minus sign.
- For reciprocal roots, reverse coefficients.
- If one root is given, substitute it to find the parameter.

8. Polynomials and Remainder Theorem

Cubic Roots

For:

$$ax^3 + bx^2 + cx + d = 0$$

roots α, β, γ:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$
$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$
$$\alpha\beta\gamma = -\frac{d}{a}$$

Remainder Theorem

Remainder when $f(x)$ is divided by $x - a$:

$$f(a)$$

Factor Theorem

$$f(a) = 0 \Rightarrow (x - a) \text{ is a factor}$$

Linear Divisor

If divisor is:

$$ax + b = 0$$

then:

$$x = -\frac{b}{a}$$

Remainder:

$$f\left(-\frac{b}{a}\right)$$

Identities

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$
$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

Tricks

- Remainder by $x - a$: substitute $x = a$.
- Remainder by $x + 2$: substitute $x = -2$.
- Remainder by $2x - 3$: substitute $x = 3/2$.
- Do not use long division unless substitution is impossible.

9. Partial Fractions

Linear Distinct Factors

$$\frac{px + q}{(x - a)(x - b)} = \frac{A}{x - a} + \frac{B}{x - b}$$

Repeated Linear Factor

$$\frac{px + q}{(x - a)^2} = \frac{A}{x - a} + \frac{B}{(x - a)^2}$$

Irreducible Quadratic Factor

$$\frac{px + q}{(x - a)(x^2 + bx + c)} = \frac{A}{x - a} + \frac{Bx + C}{x^2 + bx + c}$$

Cover-Up Method

For:

$$\frac{3x+5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Set $x = -1$ for A, set $x = -2$ for B.

Tricks

- Use zero of denominator factor to isolate coefficient.
- Repeated factor needs terms for each power.
- Quadratic denominator gets linear numerator $Bx + C$.
- Avoid full expansion if cover-up works.

10. Sequences and Series

Arithmetic Progression

$$a_n = a + (n-1)d$$
$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$S_n = \frac{n}{2}(a+l)$$

Common Difference

$$d = a_{n+1} - a_n$$

Arithmetic Mean

$$AM = \frac{a+b}{2}$$

Geometric Progression

$$a_n = ar^{n-1}$$
$$S_n = \frac{a(r^n-1)}{r-1}, \qquad r \neq 1$$
$$S_n = \frac{a(1-r^n)}{1-r}$$

Infinite GP

$$S_\infty = \frac{a}{1-r}, \qquad |r| < 1$$

Geometric and Harmonic Mean

$$GM = \sqrt{ab}$$
$$HM = \frac{2ab}{a+b}$$
$$AM \geq GM \geq HM$$

Useful Sums

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$
$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$
$$1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$
$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

Tricks

- AP means common difference.
- GP means common ratio.
- If a, x, b are in GP, then $x^2 = ab$.
- Sum of first n odd numbers is n^2.
- Use $S_n = \frac{n}{2}(a+l)$ when first and last terms are known.

11. Permutation and Combination

Factorial

$$n! = n(n-1)(n-2)\cdots 1$$
$$0! = 1$$

Permutation

Arrangement of r objects from n:

$$^nP_r = \frac{n!}{(n-r)!}$$

Combination

Selection of r objects from n:

$$^nC_r = \frac{n!}{r!(n-r)!}$$
$$^nC_r = {^nC_{n-r}}$$

Arrangements with Repetition

If total objects n, repeated groups p, q, r:

$$\frac{n!}{p!q!r!}$$

Circular Permutation

$$(n-1)!$$

Together / Not Together

Together: bundle the group first.
Not together:

$$\text{not together} = \text{total} - \text{together}$$

Tricks

- Arrangement = permutation.
- Selection = combination.
- Without repetition: choices decrease.
- With repetition: choices remain same.
- "All girls together" means bundle the girls first.

12. Probability

Basic Probability

$$P(E) = \frac{n(E)}{n(S)}$$

$$0 \le P(E) \le 1$$

$$P(S) = 1, \qquad P(\varnothing) = 0$$

Complement

$$P(E') = 1 - P(E)$$

Union

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive

$$P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B)$$

Independent Events

$$P(A \cap B) = P(A)P(B)$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Binomial Probability

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

where:

$$q = 1 - p$$

Tricks

- "At least one" means $1 - P(\text{none})$.
- Two dice: total outcomes $= 36$.
- Coin tossed n times: total outcomes $= 2^n$.
- Standard cards: total outcomes $= 52$.
- "Or" means union; subtract overlap.

13. Binomial Theorem

Expansion

$$(a+b)^n = \sum_{r=0}^{n} {}^nC_r a^{n-r} b^r$$

General Term

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Middle Term

If n is even:

$$\text{middle term} = T_{\frac{n}{2}+1}$$

If n is odd:

$$\text{middle terms} = T_{\frac{n+1}{2}},\ T_{\frac{n+3}{2}}$$

Coefficient of x^k

Use general term and equate power of x to k.

Sum of Coefficients

For polynomial $f(x)$:

$$\text{sum of coefficients} = f(1)$$

Alternating sum:

$$f(-1)$$

Constant Term

Set exponent of x equal to zero.

Tricks

- Coefficient sum: put $x = 1$.
- Constant term: power of x is zero.
- Middle term depends on even/odd n.
- In $(1+x)^n$, coefficient of x^r is ${}^nC_r$.

14. Trigonometry Basics

Ratios

$$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{\text{base}}{\text{hypotenuse}}$$

$$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$$

$$\cot\theta = \frac{1}{\tan\theta}, \qquad \sec\theta = \frac{1}{\cos\theta}, \qquad \csc\theta = \frac{1}{\sin\theta}$$

Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Special Angles

$$\sin 0 = 0, \quad \sin 30^\circ = \frac{1}{2}, \quad \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \sin 90^\circ = 1$$

$$\cos 0 = 1, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 60^\circ = \frac{1}{2}, \quad \cos 90^\circ = 0$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}, \quad \tan 45^\circ = 1, \quad \tan 60^\circ = \sqrt{3}$$

Quadrants

ASTC rule:
Quadrant I: all positive.
Quadrant II: sine positive.
Quadrant III: tangent positive.
Quadrant IV: cosine positive.

Tricks

- If $\sin\theta = 3/5$, use 3-4-5 triangle.
- If $\tan\theta = 3/4$, then $\sin\theta = 3/5, \cos\theta = 4/5$.
- Acute angle means all ratios positive.
- Never ignore quadrant in sign questions.

15. Trigonometric Identities

Compound Angles

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$
$$\sin(A-B) = \sin A\cos B - \cos A\sin B$$
$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$
$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$
$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$
$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$$

Double Angles

$$\sin 2A = 2\sin A\cos A$$
$$\cos 2A = \cos^2 A - \sin^2 A$$
$$\cos 2A = 1 - 2\sin^2 A$$
$$\cos 2A = 2\cos^2 A - 1$$
$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

Half Angles

$$\sin^2\frac{A}{2} = \frac{1-\cos A}{2}$$
$$\cos^2\frac{A}{2} = \frac{1+\cos A}{2}$$
$$\tan^2\frac{A}{2} = \frac{1-\cos A}{1+\cos A}$$

Product-to-Sum

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$
$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$
$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

Sum-to-Product

$$\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$$
$$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$$
$$\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$$
$$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$$

Maximum / Minimum

$$a\sin x + b\cos x = R\sin(x+\alpha)$$
$$R = \sqrt{a^2+b^2}$$
$$\max(a\sin x + b\cos x) = \sqrt{a^2+b^2}$$
$$\min(a\sin x + b\cos x) = -\sqrt{a^2+b^2}$$

Tricks

- $\sin 75^\circ\cos 15^\circ - \cos 75^\circ\sin 15^\circ = \sin 60^\circ$.
- For $a\sin x + b\cos x$, use $\sqrt{a^2+b^2}$.
- Rewrite $\cos^2 x - \sin^2 x$ as $\cos 2x$.
- $\tan A + \cot A = 2$ gives $\tan A = 1$ for acute A.

16. Trigonometric Equations and Graphs

Periods

$$\sin x, \cos x: \quad 2\pi$$
$$\tan x, \cot x: \quad \pi$$
$$\sin(kx), \cos(kx): \quad \frac{2\pi}{|k|}$$
$$\tan(kx): \quad \frac{\pi}{|k|}$$

Basic Solutions

$$\sin x = 0 \Rightarrow x = n\pi$$
$$\cos x = 0 \Rightarrow x = \frac{\pi}{2} + n\pi$$
$$\tan x = 0 \Rightarrow x = n\pi$$

Range

$$-1 \le \sin x \le 1$$
$$-1 \le \cos x \le 1$$
$$\tan x \in \mathbb{R}$$

Tricks

- Count solutions only in the given interval.
- Period of $\tan(2x)$ is $\pi/2$.
- If $\sin A = \cos A$ and A is acute, then $A = 45^\circ$.
- For $\sin x = 0$ in $0 \le x \le 2\pi$, solutions are $0, \pi, 2\pi$.

17. Inverse Trigonometric Functions

Principal Values

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$
$$\cos^{-1} x \in [0, \pi]$$
$$\tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Basic Identities

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$
$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

Tangent Inverse

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

with quadrant adjustment when needed.

Tricks

- Inverse trig answers depend on principal branch.
- $\sin^{-1}(\sin x) \neq x$ always.
- Use $\sin^{-1} x + \cos^{-1} x = \pi/2$ for fast simplification.

18. Limits

Standard Limits

$$\lim_{x\to 0} \frac{\sin x}{x} = 1$$
$$\lim_{x\to 0} \frac{\sin ax}{x} = a$$
$$\lim_{x\to 0} \frac{\tan x}{x} = 1$$
$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$$
$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1$$
$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$$

Algebraic Limits

$$\lim_{x\to a} \frac{x^2 - a^2}{x-a} = \lim_{x\to a} \frac{(x-a)(x+a)}{x-a} = 2a$$

Rationalization:

$$\frac{\sqrt{x}-\sqrt{a}}{x-a} \cdot \frac{\sqrt{x}+\sqrt{a}}{\sqrt{x}+\sqrt{a}}$$

Continuity

$$f \text{ continuous at } x = a \iff \lim_{x\to a} f(x) = f(a)$$

Tricks

- For 0/0, factor or rationalize.
- For $\sin ax/x$, answer is a.
- Never cancel terms across addition unless factored.
- Substitute only after removing zero denominator.

19. Differentiation

Basic Derivatives

$$\frac{d}{dx} c = 0$$
$$\frac{d}{dx} x^n = nx^{n-1}$$
$$\frac{d}{dx} \sin x = \cos x$$
$$\frac{d}{dx} \cos x = -\sin x$$
$$\frac{d}{dx} \tan x = \sec^2 x$$
$$\frac{d}{dx} \cot x = -\csc^2 x$$
$$\frac{d}{dx} \sec x = \sec x \tan x$$
$$\frac{d}{dx} \csc x = -\csc x \cot x$$
$$\frac{d}{dx} e^x = e^x$$
$$\frac{d}{dx} e^{ax} = ae^{ax}$$
$$\frac{d}{dx} a^x = a^x \ln a$$
$$\frac{d}{dx} \ln x = \frac{1}{x}$$
$$\frac{d}{dx} \log_a x = \frac{1}{x \ln a}$$

Rules

Product:

$$(uv)' = u'v + uv'$$

Quotient:

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Chain:

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

Composite Forms

$$\frac{d}{dx}(u^n) = nu^{n-1}u'$$
$$\frac{d}{dx} \ln u = \frac{u'}{u}$$
$$\frac{d}{dx} e^u = e^u u'$$
$$\frac{d}{dx} \sin u = \cos u \cdot u'$$
$$\frac{d}{dx} \cos u = -\sin u \cdot u'$$

Second Derivative

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Slope, Tangent, Normal

$$m = \frac{dy}{dx}$$

$$m = f'(a) \text{ at } x = a$$

Tangent:

$$y - y_1 = m(x - x_1)$$

Normal slope:

$$m_n = -\frac{1}{m}$$

Increasing / Decreasing

$$f'(x) > 0 \Rightarrow f \text{ increasing}$$

$$f'(x) < 0 \Rightarrow f \text{ decreasing}$$

Maxima / Minima

$$f'(x) = 0$$

$$f''(x) < 0 \Rightarrow \text{maximum}$$

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Tricks

- Derivative means slope.
- Chain rule is the most common trap.
- For $x^2 \sin x$, use product rule.
- $\frac{d}{dx} \ln(x^2 + 1) = \frac{2x}{x^2+1}$.
- $\frac{d}{dx} e^{3x} = 3e^{3x}$.

20. Integration

Basic Integrals

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \qquad n \neq -1$$

$$\int \frac{1}{x} \, dx = \ln|x| + C$$

$$\int e^x \, dx = e^x + C$$

$$\int e^{ax} \, dx = \frac{e^{ax}}{a} + C$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C$$

$$\int \sin ax \, dx = -\frac{\cos ax}{a} + C$$

$$\int \cos ax \, dx = \frac{\sin ax}{a} + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

Substitution

If:

$$\int f(g(x))g'(x) \, dx$$

let:

$$u = g(x), \qquad du = g'(x)dx$$

Example:

$$\int 2x \cos(x^2) \, dx$$

Let:

$$u = x^2, \qquad du = 2x \, dx$$

$$= \int \cos u \, du = \sin u + C = \sin(x^2) + C$$

Definite Integral

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

Area Under Curve

$$A = \int_a^b y \, dx$$

Tricks

- Indefinite integral needs $+C$.
- Definite integral does not need $+C$.
- If derivative of inside is present, use substitution.
- $\int e^{2x} dx = \frac{1}{2} e^{2x} + C$.
- $\int \frac{1}{x \ln x} dx = \ln|\ln x| + C$.

21. Analytical Geometry: Straight Lines

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Slope

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Line Forms

$$y = mx + c$$
$$y - y_1 = m(x - x_1)$$
$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$
$$\frac{x}{a} + \frac{y}{b} = 1$$
$$Ax + By + C = 0$$

Parallel and Perpendicular

$$m_1 = m_2$$
$$m_1 m_2 = -1$$

Distance from Point to Line

For point (x_1, y_1) and line $Ax + By + C = 0$:

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Tricks

- Intercepts a, b give $\frac{x}{a} + \frac{y}{b} = 1$.
- Vertical line has undefined slope.
- Horizontal line has slope zero.
- If slope $= -1$ through $(2, 3)$, then $y - 3 = -(x - 2)$.

22. Circle

Standard Form

$$(x - h)^2 + (y - k)^2 = r^2$$

Center:

$$(h, k)$$

Radius:

$$r$$

General Form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Center:

$$(-g, -f)$$

Radius:

$$r = \sqrt{g^2 + f^2 - c}$$

Diameter Form

If $A(x_1, y_1)$, $B(x_2, y_2)$ are ends of diameter:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Tangent

For:

$$x^2 + y^2 = r^2$$

tangent at (x_1, y_1):

$$xx_1 + yy_1 = r^2$$

Tricks

- Complete square for center and radius.
- In $x^2 + y^2 - 6x + 8y = 0$, center is $(3, -4)$, radius 5.
- General form uses $2g, 2f$, not g, f.

23. Conic Sections

Parabola

Right-opening:

$$y^2 = 4ax$$

Focus:

$$(a, 0)$$

Directrix:

$$x = -a$$

Left-opening:

$$y^2 = -4ax$$

Focus:

$$(-a, 0)$$

Upward:

$$x^2 = 4ay$$

Focus:

$$(0, a)$$

Downward:

$$x^2 = -4ay$$

Focus:

$$(0, -a)$$

Eccentricity:

$$e = 1$$

Ellipse

Horizontal:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b$$

Major axis:

$$2a$$

Minor axis:

$$2b$$

Foci:

$$(\pm c, 0)$$
$$c^2 = a^2 - b^2$$
$$e = \frac{c}{a}$$

Vertical:

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

Foci:

$$(0, \pm c)$$

Hyperbola

Horizontal:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Transverse axis:

$$2a$$

Conjugate axis:

$$2b$$

Foci:

$$(\pm c, 0)$$

$$c^2 = a^2 + b^2$$

$$e = \frac{c}{a} > 1$$

Vertical:

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

Foci:

$$(0, \pm c)$$

Tricks

- For $y^2 = 12x$, compare with $y^2 = 4ax$: $a = 3$, focus $(3, 0)$.
- Parabola eccentricity is always 1.
- Ellipse: $c^2 = a^2 - b^2$.
- Hyperbola: $c^2 = a^2 + b^2$.
- Major/transverse axis length is $2a$, not a.

24. Linear Inequalities and Linear Programming

Linear Inequality

$$ax + by \le c$$

Boundary line:

$$ax + by = c$$

Feasible Region

The feasible region is the common shaded region satisfying all constraints.

Corner Point Method

For:

$$Z = ax + by$$

evaluate Z at all vertices.
Maximum or minimum occurs at a corner point.

Common Constraints

$$x \ge 0, \qquad y \ge 0$$

mean first quadrant.

Tricks

- Replace inequality by equality to draw boundary.
- Test $(0, 0)$ to decide side, if not on boundary.
- Maximum/minimum occurs at a vertex.
- In MCQs, list vertices and substitute quickly into Z.

25. Vectors

Vector Form

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

Magnitude

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Unit Vector

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

Dot Product

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Perpendicular:

$$\vec{a} \cdot \vec{b} = 0$$

Cross Product

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$$

$$\hat{i} \times \hat{j} = \hat{k}$$

$$\hat{j} \times \hat{k} = \hat{i}$$

$$\hat{k} \times \hat{i} = \hat{j}$$

Reverse order gives negative:

$$\hat{j} \times \hat{i} = -\hat{k}$$

Determinant Form

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Angle Between Vectors

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

Tricks

- Dot product zero means perpendicular.
- Cross product zero means parallel.
- Magnitude cannot be negative.
- $\hat{i} \times \hat{j} = \hat{k}$.
- Dot product gives scalar; cross product gives vector.

26. Three-Dimensional Geometry

Distance in 3D

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Midpoint in 3D

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Direction Ratios

For:

$$\vec{a} = a\hat{i} + b\hat{j} + c\hat{k}$$

direction ratios are:

$$a, b, c$$

Direction Cosines

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}$$
$$m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}$$
$$n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$
$$l^2 + m^2 + n^2 = 1$$

Equation of Line

Vector form:

$$\vec{r} = \vec{a} + \lambda\vec{b}$$

Symmetric form:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Equation of Plane

$$ax + by + cz = d$$

Normal vector:

$$(a, b, c)$$

Tricks

- Direction cosines always satisfy $l^2 + m^2 + n^2 = 1$.
- Plane coefficients give normal vector.
- For distance, square all coordinate differences.

27. Variation

Direct Variation

$$A \propto B$$
$$A = kB$$

If:

$$A = A_1 \text{ when } B = B_1$$

then:

$$k = \frac{A_1}{B_1}$$

Inverse Variation

$$A \propto \frac{1}{B}$$
$$A = \frac{k}{B}$$
$$AB = k$$

Joint Variation

$$A \propto BC$$
$$A = kBC$$

Tricks

- Direct variation: ratio stays constant.
- Inverse variation: product stays constant.
- Always find k before substituting new values.

28. Quick MCQ Elimination Rules

Universal Math Traps

- Check sign first.
- Check domain restrictions.
- Check whether answer must be positive.
- Check quadrant in trigonometry.
- For determinant, use $ad - bc$.
- For roots, sum has minus sign.
- For functions, composition order matters.
- For conics, compare with standard form first.
- For probability, distinguish "and" from "or."
- For permutation/combination, distinguish arrangement from selection.

Fast Attempt Method

1. Identify the topic.
2. Write the formula.
3. Apply sign/domain/unit restriction.
4. Substitute values.
5. Compare with options.
6. Mark the trap in one line.

Most Repeated One-Line Tricks

- i^n: reduce n modulo 4.
- ω^n: reduce n modulo 3.
- $|x - a| < b$: interval is $(a - b, a + b)$.
- Singular matrix: determinant zero.
- Equal roots: discriminant zero.
- Remainder by $x - a$: put $x = a$.
- Coefficient sum: put $x = 1$.
- Constant term: exponent of x equals zero.
- $a \sin x + b \cos x$: maximum is $\sqrt{a^2 + b^2}$.
- Perpendicular vectors: dot product zero.